

KARHIV, Ukraine

WMO: 343000

Lat: 49.9281N

Lon: 36.2825E

Elev: 509

StdP: 14.43

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	2.3	-9.4	3.4	-2.1	-3.9	4.5	3.0	23.0	22.1	20.9	22.4	7.1	60	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	91.4	67.0	87.8	66.1	84.5	65.3	70.0	83.4	68.7	81.5	67.5	79.8	9.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	97.7	74.5	64.5	92.6	73.7	63.0	87.7	72.7	34.3	83.7	33.2	81.3	32.2	79.6	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.9	18.6	16.7	DB	-8.1	95.6	6.8	4.1	-13.0	98.6	-16.9	101.0	-20.7	103.3	-25.6	106.3
(5)				WB	-8.6	72.5	6.7	1.3	-13.4	73.4	-17.4	74.2	-21.1	74.9	-26.0	75.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	23.2	25.1	34.6	49.8	61.2	67.8	71.7	70.0	59.5	46.9	35.8	27.0	
	DBStd	19.37	10.54	10.62	8.18	7.75	7.98	6.77	5.96	6.79	7.61	8.29	8.89	10.12	
	HDD50	3426	831	698	478	96	9	0	0	0	11	159	431	713	
	HDD65	6879	1296	1118	942	457	169	48	10	26	196	563	876	1178	
	CDD50	2638	0	0	1	90	357	534	671	620	297	63	5	0	
	CDD65	615	0	0	0	1	52	132	216	181	32	1	0	0	
	CDH74	5775	0	0	0	25	493	1131	1989	1826	301	10	0	0	
	CDH80	2007	0	0	0	3	122	335	738	737	71	1	0	0	
Wind	WSAvg	8.3	9.1	9.3	9.5	8.8	7.8	7.6	7.0	7.0	7.5	7.7	8.8	9.4	
Precipitation	PrecAvg	21.1	1.6	1.3	1.3	1.3	1.9	2.2	2.4	1.9	1.9	1.7	1.7	1.7	
	PrecMax	29.4	5.0	2.9	3.8	3.9	5.7	5.4	6.2	6.3	4.8	4.6	5.5	5.1	
	PrecMin	12.7	0.1	0.1	0.1	0.1	0.0	0.3	0.2	0.0	0.1	0.0	0.2	0.2	
	PrecStd	4.1	1.0	0.7	0.8	0.9	1.2	1.3	1.4	1.4	1.3	1.1	1.2	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	46.4	60.6	77.4	87.9	91.2	95.7	97.9	86.4	74.2	59.3	48.6	
		MCWB	42.3	43.1	48.0	56.8	63.5	67.0	68.0	66.9	61.6	58.1	53.8	46.7	
	2%	DB	40.3	42.6	55.1	71.5	83.2	86.9	91.3	91.9	81.3	68.2	55.1	44.9	
		MCWB	38.8	40.4	44.9	54.6	62.5	66.3	67.7	66.4	60.1	56.3	51.5	43.6	
	5%	DB	37.6	39.2	50.0	67.6	79.7	84.0	87.6	87.7	77.1	63.1	51.4	41.7	
		MCWB	36.5	37.1	42.7	52.9	61.4	65.5	67.2	65.6	59.7	54.6	48.2	40.6	
	10%	DB	35.5	37.0	45.8	63.3	75.5	80.7	84.1	83.8	73.0	58.9	47.2	38.9	
		MCWB	34.4	35.4	40.6	50.8	60.2	64.5	66.3	64.7	58.4	53.1	45.0	37.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	43.4	50.1	59.1	67.7	70.8	71.7	71.3	66.3	61.5	55.4	47.1
			MCDB	43.6	45.0	57.7	73.1	80.3	83.4	85.5	85.5	78.4	69.0	58.9	48.3
2%		WB	39.0	40.4	46.3	56.4	64.9	69.1	70.3	69.2	63.7	58.4	51.9	43.6	
		MCDB	39.7	42.0	52.2	67.5	77.8	81.0	84.0	83.2	74.4	65.2	54.6	44.6	
5%		WB	36.5	37.4	43.4	54.5	63.2	67.6	69.1	67.6	61.8	55.8	48.5	40.5	
		MCDB	37.3	38.5	48.4	64.0	75.0	79.2	82.2	81.4	71.4	61.6	50.5	41.3	
10%		WB	34.2	35.3	41.0	52.5	61.5	66.2	67.9	66.2	59.8	53.5	45.1	37.5	
		MCDB	35.0	36.4	45.1	61.2	72.4	77.4	79.9	79.5	69.2	58.7	46.7	38.3	
Mean Daily Temperature Range	5% DB	MDBR	8.7	10.2	12.3	17.3	18.9	18.0	18.4	19.7	17.9	14.4	9.1	8.0	
		MCDBR	7.0	10.0	19.9	23.6	23.5	21.7	22.8	25.1	24.4	21.1	13.7	8.7	
		MCWBR	6.6	8.6	12.1	11.3	9.5	8.4	7.2	7.8	10.1	11.7	10.0	8.0	
	5% WB	MCDBR	6.6	9.1	17.2	19.9	20.5	19.3	19.5	21.0	19.9	17.9	11.9	8.3	
		MCWBR	6.6	8.3	11.6	10.7	9.3	8.5	7.6	8.4	9.8	11.5	9.8	8.0	
Clear-Sky Solar Irradiance	taub	0.317	0.348	0.379	0.456	0.412	0.402	0.433	0.434	0.403	0.367	0.348	0.321		
	taud	2.150	2.080	2.149	2.056	2.270	2.319	2.243	2.227	2.282	2.347	2.290	2.226		
	Ebn at Noon	218	241	258	249	269	272	261	254	249	237	208	200		
	Edh at Noon	25	35	39	48	41	39	42	40	34	27	23	21		
All-Sky Solar Radiation	RadAvg	299	594	972	1351	1818	1942	1901	1686	1144	645	280	187		
	RadStd	61	108	149	133	143	159	126	131	139	103	59	38		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.51	N/A	N/A	N/A	N/A	N/A	-315	-425	+227	+103
(47)	Regional (7 neighbors)	+1.37	N/A	N/A	+1.78	N/A	N/A	-284	-394	+205	+119

Nomenclature: See separate page